Individual Ali

Design

Hardware Used

- Arduino UNO Microcontroller
- Humidity Sensor
- LDR Potentiometer
- LCD Display (20x4)
- Dimmable Lamp
- Fan
- Water Spray
- LED

Software Used

- Arduino (C Programming)
- Proteus (Simulation)

System Description

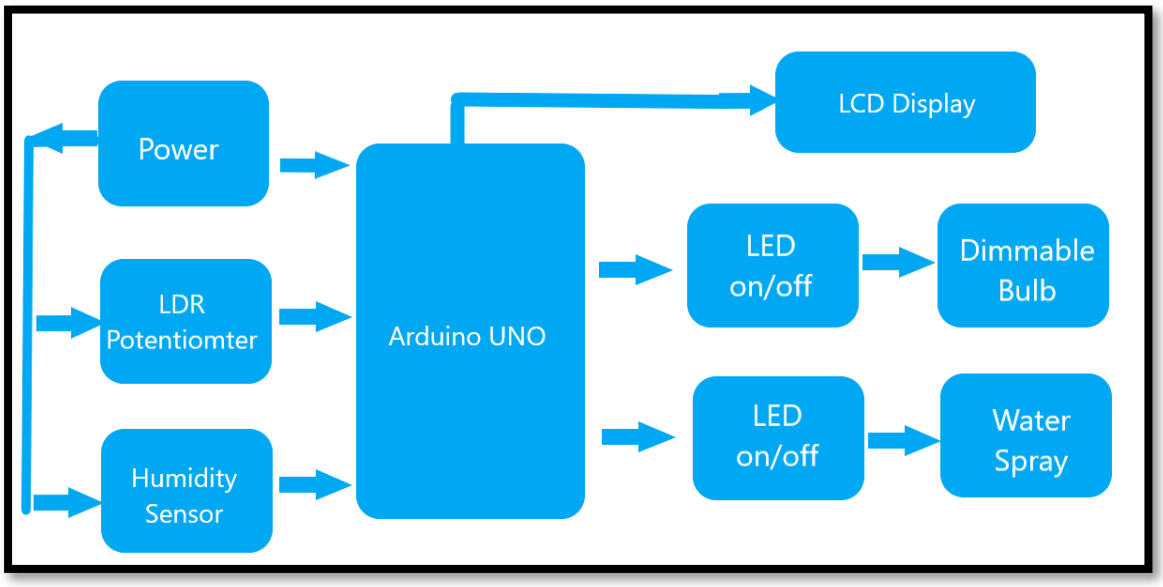


Figure 1 : Block Diagram (Process)

- **Light Sensor**: Senses the light intensity, resistance increases as light intensity increases and vice versa.
- **Humidity Sensor:** Senses the humidity level in the near environment.
- **Arduino UNO**: Is an 8-bit ATmega328P microcontroller-based microcontroller module. Along with ATmega328P, it consists of other components to assist the microcontroller such as crystal oscillator, serial communication, voltage regulator etc. Arduino Uno has 14 digital input / output pins (6 of which can be used as PWM outputs) , 6 analog input pins, a USB connection , A Power barrel jack, ICSP header and a reset button.
- **LCD (20x4)**: It is used to display the current readings on the screen.
- **Variable Lamp:** It is used to adjust the light intensity when required.
- **LED:** Used to indicate the status of the measured values if they are higher or lesser than optimal level.

Function

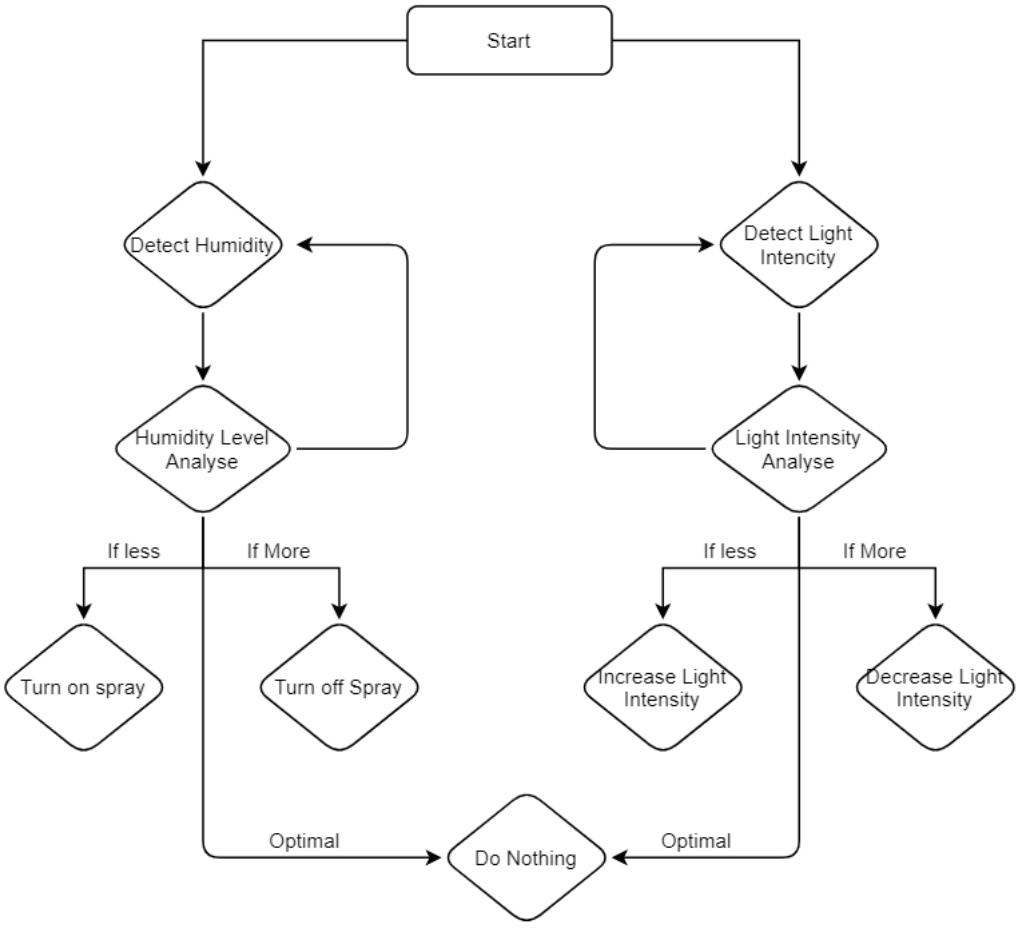


Figure 2 : Flow Chart of System

The project is designed to function as an automatic plant monitoring system which turns the spray/light ON/OFF depending upon the humidity level and light intensity on the plant. The project uses Arduino UNO microcontroller which is programmed to receive the input signal of varying humidity and light condition of the plant through the sensing arrangement. This is achieved by using a humidity sensor and LDR as detector which acts as interface between the sensing arrangement and the microcontroller. An LCD display is also interfaced to the microcontroller to display status of the humidity level and light intensity. The sensing arrangement is made by using the sensors near the plant.

The microcontroller is programmed according to the plants desired growing condition keeping the humidity and light variables at optimal. When the sensor detects that the humidity and light intensity measurement it checks with the system whether the readings are above or below optimal level and responds accordingly.

Programming

```
const int ledPin13 = 13;
const int ledPin12 = 12;
const int ledPin11 = 11;
const int ledPin10 = 10;
const int ledPin9 = 9;
const int ledPin8 = 8;
const int ledPin7 = 7;
const int ledPin6 = 6;
const int ledPin5 = 5;
const int ledPin4 = 4;
const int ledPin3 = 3;
const int ledPin2 = 2;
const int ledPin1 = 1;
const int ledPin0 = 0;
const int ldrPin = A0;
int soilsensor_Pin = A3;
int soilstatus;
int waterPin = A1;
int waterstatus;

#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 20, 4);

#include <DHT.h>
#define DHTPin 2
#define DHTTYPE DHT11
DHT dht(DHTPin, DHTTYPE);
```

In the image above the program for the arduino is done, firs the variables are first defined with their designated pin numbers. The libraries for the sensors are included as well as for the lcd display.

```
Serial.begin(9600);
dht.begin();
pinMode(ledPin13, OUTPUT);
pinMode(ledPin12, OUTPUT);
pinMode(ledPin11, OUTPUT);
pinMode(ledPin10, OUTPUT);
pinMode(ledPin9, OUTPUT);
pinMode(ledPin8, OUTPUT);
pinMode(ledPin7, OUTPUT);
pinMode(ledPin6, OUTPUT);
pinMode(ledPin5, OUTPUT);
pinMode(ledPin4, OUTPUT);
pinMode(ledPin3, OUTPUT);
pinMode(ledPin2, OUTPUT);
pinMode(ldrPin, INPUT);
pinMode(soilstatus, OUTPUT);
```

This is where the main code begins first we use Serial.begin(9600) to set the data rate in bits per second (baud) for serial data transmission. Then we define which pin is input and output. In this case led 13-2 are all outputs as they will be the response to the programming. And two inputs which is the LDR and humidity readings.

```
lcd.begin();
lcd.print("Weed Monitor");
lcd.setCursor (0, 1);
lcd.print("MSES <3");
lcd.setCursor (0, 2);
lcd.print("By");
lcd.setCursor (0, 3);
lcd.print("Ali  Nabz  Shim");
delay(2000);
```

In this section we initialise the led screen. Since the led is 20x4, we define on the first line the title “Weed Monitor” and then the rest of the line are defined. Each line has to given a separate loaction as they start from line 0-3 and this sentense last for 2 seconds on the lcd screen.

```
void loop() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();
  int ldrStatus = analogRead(ldrPin);
  soilpin = analogRead(sensor_pin);
  soilpin = map(soilpin, 550, 10, 0, 100);

  //FOR LCD & TEMP//

  lcd.begin();
  lcd.backlight();
  lcd.clear();

  lcd.setCursor (0, 0);
  lcd.print("Light Int: ");
  lcd.print(ldrStatus);
  Serial.print(ldrStatus);

  lcd.setCursor (0, 2);
  lcd.print("Humidity: ");
  lcd.print(h);
  lcd.print("%");
  Serial.print(", ");
  Serial.print(h);
```

In this section the readings for Light intensity and humidity are given variables “ldrStatus” and “h” respectively. Then the readings are displayed on the lcd screen and after every 2 seconds the values are updated.

```
if (ldrStatus > 200) {

  digitalWrite(ledPin13, HIGH);
  digitalWrite(ledPin11, LOW);
  lcd.setCursor (0, 1);
  lcd.print("ITS DARK");
  lcd.setCursor (0, 2);
  lcd.print("LIGHT ON");
  delay (2000);

} else if (ldrStatus < 150) {

  digitalWrite(ledPin13, LOW);
  digitalWrite(ledPin12, HIGH);
  lcd.setCursor (0, 1);
  lcd.print("ITS BRIGHT");
  lcd.setCursor (0, 2);
  lcd.print("LIGHT OFF");
  delay (2000);
}
else if (150 > ldrStatus < 200) {

  digitalWrite(ledPin13, LOW);
  digitalWrite(ledPin12, LOW);
  lcd.setCursor (0, 1);
  lcd.print("ITS OPTIMAL");
  lcd.setCursor (0, 2);
  lcd.print("NO ACTION");
  delay (2000);
}
```

In this section the response to the readings are set using an if statement. If the value of the LDR is above 200 meaning the light intensity is low so the ledpin 13 will be lit up showing the indication and the lamp will turned on. If the value of the LDR is below 150 meaning the light intensity is higher then the ledpin 12 will be lit up and the Lamp tuned off. If the LDR readinh is within 150-200 then no change will occur

```
if (h > 60) {

  digitalWrite(ledPin8, HIGH);
  digitalWrite(ledPin7, LOW);
  digitalWrite(ledPin6, LOW);
  lcd.setCursor (0, 1);
  lcd.print("HIGH HUMIDITY");
  lcd.setCursor (0, 2);
  lcd.print("FAN ON");
  delay (2000);

} else if (h < 50) {

  digitalWrite(ledPin8, LOW);
  digitalWrite(ledPin7, LOW);
  digitalWrite(ledPin6, HIGH);
  lcd.setCursor (0, 1);
  lcd.print("LOW HUMIDITY");
  lcd.setCursor (0, 2);
  lcd.print("SPRAY ON");
  delay (2000);
}
else if (50 > h < 60) {

  digitalWrite(ledPin8, LOW);
  digitalWrite(ledPin7, HIGH);
  digitalWrite(ledPin6, LOW);
  lcd.setCursor (0, 1);
  lcd.print("ITS OPTIMAL");
  lcd.setCursor (0, 2);
  lcd.print("NO ACTION");
  delay (2000);
}
```

In this section the response to the humidity readings are set using an if statement. If the value of the humidity sensor is above 60 meaning the humidity is high so the ledpin 8 will be lit up showing the indication. When the led is lit up it will follow up on its response which is to turn off the water spray and turn on the fan. If the value of the humididty sensor is below 50 meaning the humidity level is lower then optimal then the ledpin 6 will be lit up and the spray tunred on and fan off. If the humidity level is within 50-60 then no change will occur and only the ledpin 7 will be lit which indicates its optimal.

Since this under voidloop this program will keep going in a loop and updating the values every second.

Enhancement

GSM

This system is linked to a communication the device is called the "GSM Module." It could be connected to Various devices such as modems, cell phones or the satellite terminal to activate the remote-control Collection of recorded data or alarm of certain data It's parameters. The connection between the two Mobile devices are made using GSM. When it comes to the humidity and light levels, sensor detects the high or low reading and it sends a signal from the sensor to the microcontroller. The microcontroller then sends a signal to the GSM Module which further sends a message to the mobile.

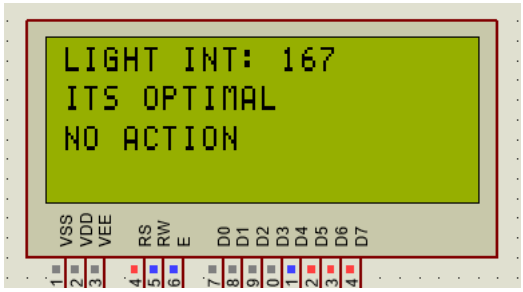
Water Sensor

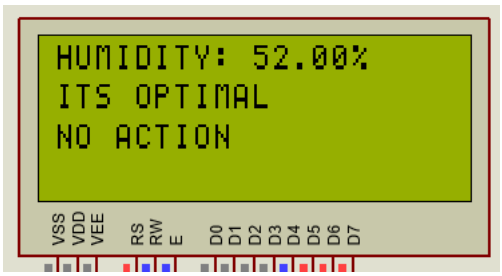
This sensor is attached to the water container that carries the water for the spray. It helps the user by sending the signal to the mobile indicating if the water level is low and it needs to be refilled.

Results

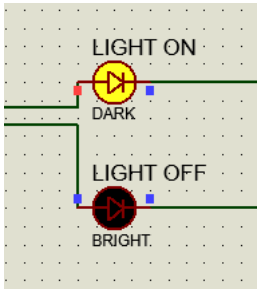
Humidity	Fan	Spray Motor	Light Intensity	Lamp
1	Off	On	123	On
15	Off	On	138	On
27	Off	Off	160	Off
53	On	Off	198	Off
64	On	Off	213	Off

Optimal Condition (Simulation)



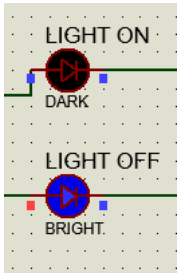


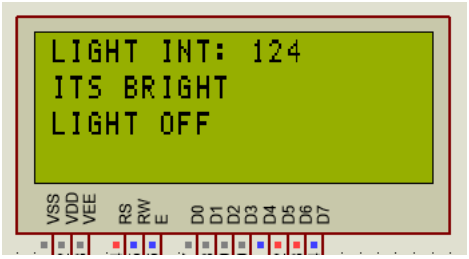
Light On (simulation



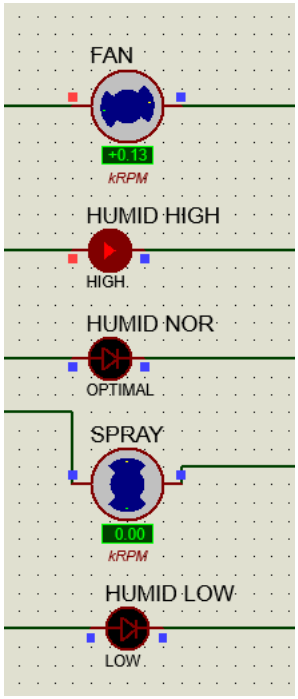


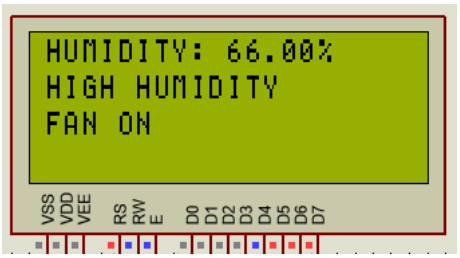
Light off (Simulation)



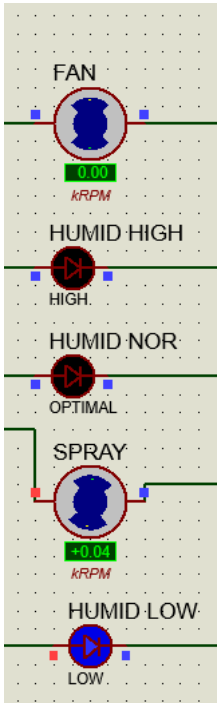


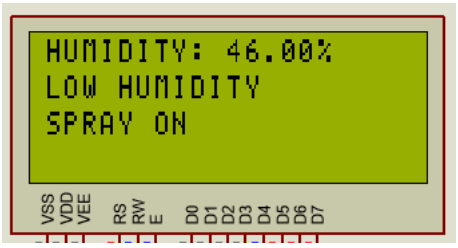
Humidity High (Simualtion





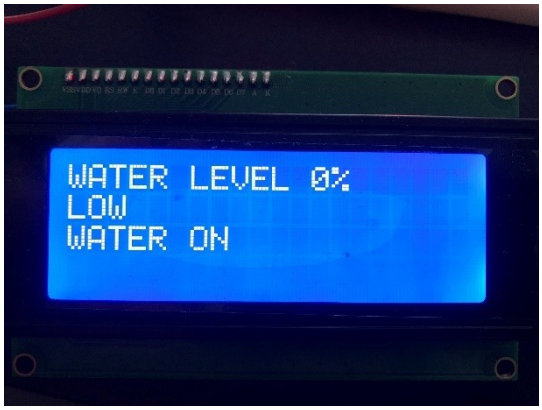
Humidity Low (Simualtion)





Real-Life Results





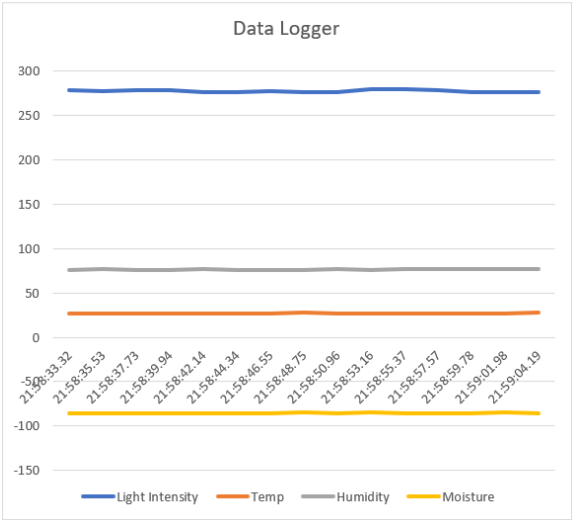


Data Record(automatic)

Time	Light Intensity	Temparature°C	Humidity	Mositure
21:59:04.19	276	28.2	77	-86

Historical Data

Time	Light Intensity	Temparature°C	Humidity	Mositure
21:58:33.32	279	28.1	76	-86
21:58:35.53	278	28.1	77	-86
21:58:37.73	279	28.1	76	-86
21:58:39.94	279	28.1	76	-86
21:58:42.14	276	28.1	77	-86
21:58:44.34	277	28.1	76	-86
21:58:46.55	278	28.1	76	-86
21:58:48.75	277	28.2	76	-85
21:58:50.96	276	28.1	77	-86
21:58:53.16	280	28.1	76	-85
21:58:55.37	280	28.1	77	-86
21:58:57.57	279	28.1	77	-86
21:58:59.78	276	28.1	77	-86
21:59:01.98	277	28.1	77	-85
21:59:04.19	276	28.2	77	-86



Discussion

The system used provides us with the readings of the light intensity and humidity level of the plant environment. These data are used to keep a track of the requirements if the field and to keep a check on the proper functioning of the system. The Automatic plant monitoring system is essentially designed for the introduction of embedded technology in the irrigation sector. This system will help farmers to reduce their pressure of work. This system will be of assistance to farmers save their precious time and produce more crops. It will certainly be helpful for farmers to improve their economic condition. The automatic operation capacity of this system requires a minimum quantity for human monitoring work and increases its contribution.

- Saves water
- Improves growth
- Discourages weeds
- Saves time
- Helps control fungal diseases
- Adaptable
- Eliminates the manual operation of opening or closing valves
- Adaption of the advanced monitoring systems and the new technologies, especially the new monitoring systems that are complex and difficult to operate manually.

- The system will be operated in night also which results in minimization of the water loss due to evaporation.
- The processes starts and stops exactly when required, thus optimizing energy requirements